Ophthalmia Neonatorum

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Introduction

- ☑ Ophthalmia neonatorum refers to any conjunctivitis occurring in the first 28 days of life.
- ☑ It can be a sight-threatening condition of the eye seen in the first 4 weeks of life. Reported incidence in India is 0.5–33%.

Etiology

Flowchart 1: Etiology of ophthalmia neonatorum.

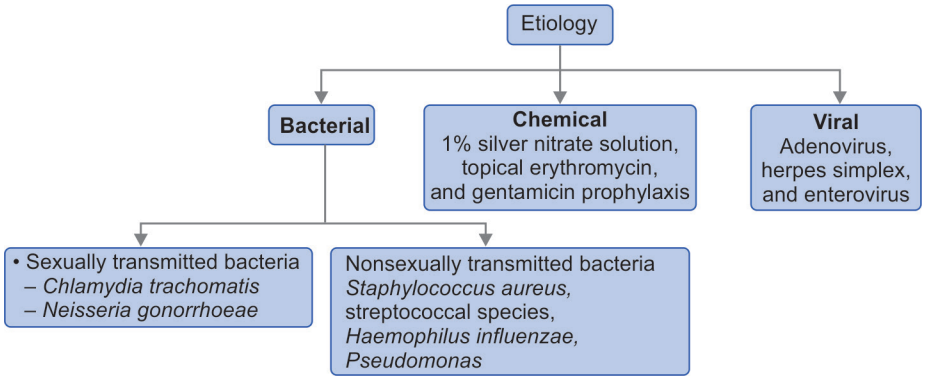

TABLE 1: Signs and symptoms of ophthalmia neonatorum.

Causative agent	***Signs and symptoms***	***Time of appearance***	***Special consideration***
Chlamydia trachomatis (chlamydial ophthalmia)	Minimal mucopurulent discharge to severe eyelid edema with copious discharge and pseudomembrane formation	5–14 days after birth	Most common cause of late onset conjunctivitis Maternal cervix or urethra serves as reservoir of the organism *Chlamydia pneumonitis* may accompany neonatal conjunctivitis
Neisseria gonorrhoeae (gonococcal ophthalmia)	Severe eyelid edema, profuse mucopurulent discharge, chemosis, corneal ulceration, and blindness, if untreated	2–5 days after birth, earlier in premature rupture of membrane	Most virulent infectious cause of ophthalmia neonatorum
Chemical	Mildly red eye with some swelling of eyelids following topical prophylaxis	6–8 hours after instillation of topical prophylaxis	Disappears spontaneously after 48 hours
Herpes simplex (herpetic keratoconjunctivitis)	Red eye, serosanguinous discharge, vesicles on lid margin or skin, and hazy cornea due to edema Presence of dendritic keratitis is pathognomonic	Appears anytime during the first 2 weeks of life	Rare cause. Can be associated with generalized herpes simplex infection
Pathogenic bacteria Coagulase-negative *Staphylococcus* Alpha-hemolytic *Streptococcus* *Haemophilus influenzae* *Streptococcus pneumoniae* *Staphylococcus aureus* *Pseudomonas* species *Escherichia coli* species	Conjunctival redness and discharge	Within 72 hours of birth	Most common cause of conjunctivitis Occurs in vaginal deliveries Premature rupture of membranes is a risk factor Midwife examination is another risk factor

Evaluation

- ☑ *Complete history*:
  - History of time of appearance of symptoms and signs
  - History of risk factors, e.g., maternal sexually transmitted disease (STD), premature rupture of membrane, prolonged delivery, midwife interference, and unclean vaginal examination.
- ☑ *Examination*:
  - Conjunctivitis (discharge, edema/erythema of lids, and conjunctival hyperemia)
  - Discharge (purulent in bacterial, watery or serosanguinous in viral, and greenish in *Pseudomonas* infection)
  - Unilateral eye involvement in *Staphylococcus aureus*, *Pseudomonas,* and viral conjunctivitis
  - Red reflex
  - Systemic examination—look for systemic sepsis, abscess, cellulitis, and disseminated infection.
- ☑ Eye discharge Gram stain and culture in Thayer Martin media and chocolate agar for bacteria, especially if gonococcal ophthalmia is suspected.
- ☑ Conjunctival scraping polymerase chain reaction (PCR), Giemsa staining, or direct fluorescent antibody staining for chlamydial ophthalmia.
- ☑ Conjunctival swab PCR for suspected herpetic infection. Blood and cerebrospinal fluid (CSF) analysis to exclude central nervous system (CNS) and disseminated disease.

Management

If there is a high degree of suspicion of ophthalmia neonatorum but lack of confirmatory test, start treatment for both chlamydial and gonococcal ophthalmia to prevent sight-threatening complications **(Table 2)**.

TABLE 2: Treatment for ophthalmia neonatorum.

Organism	*First choice*	*Adjunct*	*Special considerations*
Gonococcal	As per WHO STI guidelines Single dose of ceftriaxone injection IM 50 mg/kg/day (maximum 150 mg) Or Kanamycin 25 mg/kg (maximum 75 mg) IM single dose Or Spectinomycin 25 mg/kg (maximum 75 mg) IM single dose	Normal saline irrigation of eyes every 1–2 hours Topical antibiotic not necessary	☑ Isolate baby during first 24 hours of parental antibiotic therapy ☑ Consider treating for chlamydia due to high rate of concomitant infection ☑ Evaluate for disseminated disease ☑ Test for other STD ☑ Take ophthalmology consultation

Contd...

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Organism	***First choice***	***Adjunct***	***Special considerations***
Chlamydial	As per WHO STI guidelines Azithromycin 20 mg/kg once daily for 3 days is preferred over erythromycin 50 mg/kg/day orally in four divided doses for 2 weeks	Frequent normal saline irrigation of eyes	☑ Neonatal prophylaxis with topical antibiotics do not prevent chlamydial ophthalmia ☑ For infants born to mothers with chlamydia exposure, educate family to monitor baby for infection including pneumonia
Herpetic	Parenteral acyclovir 20 mg/kg three times a day for 14 days (21 days if disseminated or CNS disease) along with topical ganciclovir 0.15% or 1% trifluridine five times a day for 10–14 days		☑ Seek ophthalmology consultation. Keratitis, retinopathy and chorioretinitis can develop ☑ Evaluate and treat for systemic herpes ☑ Isolate baby
Coagulase-negative *Staphylococcus* Alpha-hemolytic *Streptococcus* *Haemophilus influenzae* Streptococcus *pneumoniae* *Staphylococcus aureus* *Pseudomonas* species *Escherichia coli* species	Ophthalmic drops—tobramycin or gentamicin × three to four times a day × 7 days		☑ Most common cause of conjunctivitis ☑ Occurs following vaginal delivery

(CNS: central nervous system; STD: sexually transmitted disease; STI: sexually transmitted infection)

Prevention for Ophthalmia Neonatorum

Prophylaxis

- ☑ Silver nitrate prophylaxis for gonococcal ophthalmia neonatorum has been recommended in areas where incidence of maternal gonorrhea infection is high.
- ☑ 1% silver nitrate solution (Credé's method), 0.5% erythromycin ointment, 1% tetracycline hydrochloride, chloramphenicol 1% eye ointment, or 2.5% povidone iodine (aqueous solution without alcohol) is used for prophylaxis.
- ☑ However, it is not practiced in most parts of our country; the WHO STI guidelines 2017 recommend ocular prophylaxis for prevention of chlamydia and gonococcal infection. This guideline suggests one of the above-mentioned agents for topical application to both eyes immediately after birth.
- ☑ Administration of colostrum into the eye has been shown in a small study to reduce the incidence of neonatal conjunctivitis.
- ☑ Avoid cross contamination by frequent hand washing.
- ☑ Avoid eye patching.
- ☑ Chemical conjunctivitis resolves spontaneously. May need artificial tears.

Further Reading

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Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Epistaxis in Children

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